



NATIONAL SENIOR CERTIFICATE EXAMINATION
NOVEMBER 2010

INFORMATION TECHNOLOGY: PAPER II

Time: 3 hours

120 marks

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of 7 pages. Please check that your question paper is complete.
2. This question paper is to be answered using Object Oriented Programming principles. Your program must make sensible use of methods and parameters.
3. This paper is divided into two sections. All candidates must answer both sections.
4. This paper is set in programming terms that are not specific to any particular programming language (Java/Delphi) or database (Access/MySQL). All candidates must answer all questions.
5. Make sure that you answer the questions in the manner described because marks will be awarded for your solution according to the specifications that are given in the question.
6. Only answer what is asked in each question. For example, if the question does not ask for data validation, then no marks are awarded for it, and therefore no code needs to be written.
7. If you cannot get a section of code to work, comment it out so that it will not be executed and so that you can continue with the examination. If possible, try to explain the error to aid the marker.
8. When accessing files from within your code, DO NOT use full path names of the file, as this will create problems when the program is marked on a computer other than the one you are writing on. Merely refer to the files using their names and extensions, where necessary.
9. Your programs must be coded in such a way that they will work with any data and not just the sample data supplied or any data extracts that appear in the question paper. You are advised to look at the supplied data files carefully.
10. Make sure that routines such as searches, sorts and selections are developed from first principles, and that you do not use the built-in features of a programming language for any of these routines.
11. All data structures must be defined and declared by you, the programmer. You may not use components provided within the interface to store and later retrieve data.

12. Read the whole question paper before you choose a data structure. You will find that there may be an alternative method of representing the data which will be more efficient considering the questions that are asked in the paper.
 13. You must save all your work regularly on the disk you have been given, or the disk space allocated to you for this examination.
 14. If there is a technical interruption that prevents you from writing your examination, (e.g. a power failure), when you resume writing your examinations, you will only be given the time that was remaining when the interruption began. No extra time will be given to catch up work that was not saved.
 15. Make sure that your examination number appears as a comment in every program that you code as well as on every page of hard copy that you hand in.
 16. Print a code listing of all the programs/classes that you code. Printing must be done after the examination. You will be given half an hour to print after the examination is finished. Your teacher will tell you what arrangements have been made for the printing of your work.
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SECTION A STRUCTURED QUERY LANGUAGE

The files required for this question can be found in the folder called **Question1**. Please save your answers to this question in the answer file called **SQL Answer Sheet**.

You are strongly advised to make a backup copy of the database files should you wish to revert back to the original data, for any reason.

The Gauteng and KwaZulu-Natal Traffic Departments are considering a joint initiative in traffic enforcement on the N3 highway between Johannesburg and Durban. They have captured some data of people who have committed traffic offences on the N3 highway in a database and now want some SQL queries that will summarise the data that appears in the database.

A SQL browser tool has been provided for your use. You may run this program and use it to test your SQL statements before placing your answers on the Answer Sheet.

To use the software, you simply type the query into the space provided and click the 'Execute SQL' button. The results will be displayed in the grid.

You may alternatively use any other SQL tool that will allow you to test your statement.

The database has three tables in it:

tblCars

CarRegistration	This field contains the registration number that appears on the car's number plate.
CarModel	This field contains a description of the car's make and model.
CarYear	This field contains the year in which the car was made.
OwnerID	This field contains a foreign key to associate the car with its owner in the tblOwner table.

tblOwners

OwnerID	This field contains a unique identifying number for each car owner.
OwnerLastName	This field contains the last name of the owner.
OwnerFirstName	This field contains the first name of the owner.
OwnerAge	This field contains the age of the owner on 1 January 2010.
OwnerCell	This field contains the cellphone number of the owner.

tblFines

RecordID	This field contains a unique number associated with each fine.
CarRegistration	This field contains a foreign key to associate the fine with the car that committed the offence.
FineAmount	This field contains the amount of the fine in Rands.
FineDate	This field contains the date on which the offence was committed.
FineOfficerID	This field contains the Officer Identification Number of the traffic officer that issued the fine.

- 1.1 List all the cars that were manufactured in 2010. (3)
- 1.2 List all the cars that are registered with Gauteng registrations and thus end in 'GP'. (5)
- 1.3 How many Volkswagen Polos are listed in the **tblCars** table? (4)
- 1.4 List all the cars that belong to owners who are 45 years old or older. (6)
- 1.5 How much money was collected from fines from cars that were manufactured after 2008? (7)
- 1.6 Some cars have received more than one fine. List the total amount owed by each car. (5)

Note: *You are strongly encouraged to make a backup of your database at this point. The following queries will change the data in your database. If you wish to re-test these queries, you will need to be able to restore the original data.*

- 1.7 Write an SQL statement which will change the fines table and reduce any fines that are less than 32 days old by half. (6)
- 1.8 A traffic officer, whose Officer Identification Number is TPDW40NN5, has been found guilty of accepting bribes from motorists. Because of this, all fines that this officer has issued must be deleted from the database. (4)

40 marks

SECTION B OBJECT ORIENTED PROGRAMMING

Considering the large volume of traffic that travels between Gauteng and KwaZulu-Natal each year, the Gauteng and KZN Traffic Departments are looking at potential ways to simplify their vehicle licensing process. In this section, you will be asked to investigate a possible licensing scenario. To assist in this, you are given data in the file called **licence.txt**.

This file contains the year of manufacture and licence plate numbers of a sample of vehicles chosen at random. The first five lines of this file appear below:

```
2007#DQS759GP
1990#OYN201GP
1998#ZMW463GP
1992#NV340701
2009#TPJ353GP
```

Your job is to write a program that can perform some basic tasks to summarise the data for them.

The National Government has established the standard licensing fee for a motor vehicle at R250 per year. However, the individual provinces are able to make adjustments to this amount based on criteria that they motivate to the Government. You will need to be aware of the following criteria that are in place:

Gauteng:

- All cars pay 1.2 times the national licensing fee because of the higher road utilisation in the province.
- All Gauteng licence plate numbers in the data file start with three letters, followed by three numbers and end with the letters 'GP'.
- Any licence plate number that contains vowels is considered to be a personalised licence plate number. For every vowel contained in the licence plate number, the licence fee must double. Finally, the numeric digits must be added to the total.

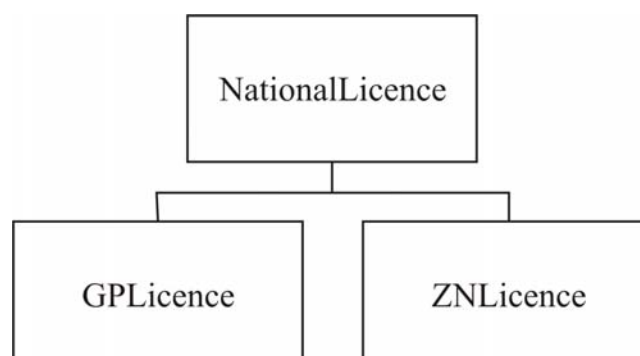
For example:

- ABC123GP would pay $(R250 * 1.2) * 2 + R123 = R723$ (one vowel)
- EAV468GP would pay $(R250 * 1.2) * 2 * 2 + R468 = R1668$ (two vowels)

KwaZulu-Natal:

- Cars that were manufactured in 1999 or before must pay R50 more than the national licensing fee because of increased environmental damage.
- All KwaZulu-Natal licence plate numbers start with two letters (the first of which is always an 'N') followed by 6 digits.
- None of the KwaZulu-Natal licence plate numbers in the data file are personalised.

In order to model these criteria, the following class design has been proposed:



2. Starting with the **NationalLicence** class:

- 2.1 Create a class called **NationalLicence**. (2)
- 2.2 Add properties in the class to store the year the car was manufactured as well as its licence number. Choose appropriate names and data types for these two properties and ensure that they cannot be accessed from outside the class. (3)
- 2.3 Create a constructor method which will initialise both properties. (4)
- 2.4 Create accessor methods for both properties. (3)
- 2.5 Create a method called **getLicenceFee** that will return the standard licensing fee of R250. (2)
- 2.6 Create a **toString** method that will return the licence plate number and the licence fee required. Separate the two properties with a **tab** character. (4)

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3. Let's consider the **GPLicence** class:

- 3.1 Create a class called **GPLicence**. This will inherit from the **NationalLicence** class. (2)
- 3.2 Create a constructor method that will initialise the two properties of the parent class. (3)

*We are not finished with the **GPLicence** class yet. We still need to add the licence fee calculation method. We will do this later on in this examination (in Question 7). You may attempt that question now, but you are encouraged to continue with the questions in the order that they have been given to you.*

[5]

4. Consider the **ZNLicence** class:

- 4.1 Create a class called **ZNLicence** which inherits from the **NationalLicence** class. (2)
- 4.2 Create a constructor method that will initialise the two properties of the parent class. (2)
- 4.3 Create a method to override the **getLicenceFee** method created in the **NationalLicence** class. This method should return the licensing fee as determined by the rules for vehicles registered in KZN. It should be able to take into account any change in the national licensing fee that the National Government might make. (5)

[9]

5. Now consider a **LicenceManager** class.
- 5.1 Create a class called **LicenceManager**. This class should have, declared as instance variables, an array of objects that could store a maximum of 200 of either **ZNLicence** or **GPLicence** objects and an appropriately named and typed variable to keep count of the total number of objects that have been instantiated in the array. (5)
- 5.2 Create a constructor method in the **LicenceManager** class that will read the data stored in the **licence.txt** file and load it into the array. As you read in each vehicle, check the licence plate to see if the last two characters are 'GP'. If this is the case, instantiate a **GPLicence** object. Otherwise, instantiate a **ZNLicence** object. (9)
- 5.3 Create a method called **totalLicenceFees** that will calculate the total licence fees owed from all of the cars. (6)
- 5.4 Create a method called **getGPLicenceList** that will return a String containing a list of all the cars from Gauteng and how much they owe. (6)
- 5.5 Create an appropriately named method that will return the **toString** value of the vehicle with the highest licence fee payable. (6)
- [32]
6. Create a simple interface that will instantiate a **LicenceManager** object and then display the output of the methods created in 5.3, 5.4 and 5.5 above. No menu or any form of user interaction is required. (6)
7. In Question 3, you created the **GPLicence** class. Until now, however, we have not implemented the algorithm to determine the licensing costs for cars from Gauteng. In the **GPLicence** class, override the **getLicenceFee** method in order to determine the licence fee payable by drivers of Gauteng vehicles. The licence fee can be returned as an integer. (10)

80 marks

Total: 120 marks